

EEE 416 (January 2025)  
Microprocessor and Embedded Systems Laboratory

**Final Project Report**

**Section: B2 Group: 08**

**IoT-Based Smart Walking Stick for Enhanced  
Mobility of the Elderly and Visually Impaired  
People.**

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**Course Instructors:**

Khairul Islam, Lecturer  
Akif Hamid, Part-Time Lecturer

Signature of Instructor: \_\_\_\_\_

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|                                                                                                      |                                                                                                   |  |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--|
| <b>Signature:</b><br>_____<br><b>Full Name: Md Mahmud Akon</b><br><b>Student ID: 2006127</b>         | <b>Signature:</b><br>_____<br><b>Full Name: Md. Fahim</b><br><b>Student ID: 2006128</b>           |  |
| <b>Signature:</b><br>_____<br><b>Full Name: Md. Marziul Haque Riad</b><br><b>Student ID: 2006129</b> | <b>Signature:</b><br>_____<br><b>Full Name: Kamrul Hasan Shanto</b><br><b>Student ID: 2006130</b> |  |

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# 1 Abstract

Mobility and safety are significant concerns for elderly and visually impaired individuals, who often face challenges such as navigating obstacles and a heightened risk of falls. Our project, titled "IoT-Based Smart Walking Stick for Enhanced Mobility of the Elderly and Visually Impaired People," presents the design and development of an intelligent walking aid to address these issues. The primary objective is to enhance user safety, confidence, and independence through the integration of modern sensor and communication technologies. The system is architected around an ESP32 microcontroller and an Arduino, which serves as the central processing unit. A suite of sensors is integrated to provide comprehensive environmental awareness and user monitoring. Ultrasonic sensors are employed for robust obstacle detection, including walls, and are specifically configured to differentiate between ascending and descending staircases. An accelerometer is used for reliable fall detection. For location tracking, a GPS module provides real-time tracking, which is crucial for emergencies. In the event of a fall or a manually triggered alert via an emergency push button, the system gives an output to an HTTP webpage via ESP that can only be accessed by pre-registered caregivers or family members. The smart stick also provides immediate feedback to the user through sound alerts. By combining obstacle detection, fall alerts, and real-time tracking into a single, cost-effective device, our project aims to offer a practical solution that significantly improves the quality of life for the elderly and visually impaired.

## 2 Introduction

For many people, getting around freely is something we take for granted. But for millions of elderly and visually impaired individuals, navigating the world can be a daily challenge filled with uncertainty. A simple walk can present risks, from unseen obstacles to the fear of a fall with no one around to help. While the traditional walking cane has been a longtime companion, it hasn't evolved to meet the modern needs for safety and connectivity. It's a tool that offers support but lacks the intelligence to actively protect its user.

Our objective is to design and build an IoT-based smart walking stick that does more than just provide physical support; it acts as a vigilant companion to enhance the safety and mobility of its user. Our smart stick is designed with several key features to provide a comprehensive safety net. It uses ultrasonic sensors to detect obstacles in the user's path, an accelerometer to instantly sense a fall, and a GPS module for real-time location tracking. In an emergency, the system can automatically send an alert message with the user's location to family members or caregivers. All this information is also integrated with a web application, giving loved one's peace of mind.

Ultimately, this project is about creating a device that empowers people, giving them the confidence to move more freely and live with greater independence. We're transforming passive aid into an active safety device that keeps users and their loved ones connected.

## 3 Design

### 3.1 Problem Formulation (PO(b))

Elderly and visually impaired individuals rely heavily on walking aid for mobility, yet the traditional cane has remained largely unchanged for centuries. This project identifies and aims to solve a set of critical, interconnected problems:

**Lack of Proactive Hazard Detection:** A standard cane offers only tactile feedback, meaning a user discovers an obstacle like a wall, a piece of furniture, or a sudden drop-off (like a downward staircase) only when they are already upon it. This reactive method of navigation is insufficient and leads to a high risk of collisions, trips, and stumbles.

**The Silent Danger of Falls:** Falls are a leading cause of serious injury among the elderly. A conventional cane offers no assistance once a fall occurs. The user is left to call for help themselves, which may be impossible if they are injured or unconscious.

**Difficulty in Emergency Situations and Disorientation:** In a moment of crisis—be it a medical emergency or simply becoming lost—a user may be unable to effectively communicate their situation or their precise location. Relying on a separate mobile phone can be difficult for those with visual impairments or during a stressful event.

**Caregiver Anxiety and Lack of Connection:** For family members and caregivers, there is constant concern for the well-being of their loved ones when they are out of sight. This creates a sense of anxiety, stemming from the inability to know if the person is safe or where they are.

Our project aims to solve this by developing a single, integrated device that addresses these shortcomings, transforming the walking stick into an intelligent companion that actively protects its user.

#### 3.1.1 Identification of Scope

This scope defines the specific features and goals we are committed to completing in this project. Our primary focus is to design and create a single, fully functional prototype of a smart walking stick. This device will be a complete system with all the core features working together. Here's a breakdown of what that includes:

- Obstacle Detection: Using ultrasonic sensors to warn the user about objects in their path.
- Fall Detection: Integrating an accelerometer (MPU6050) that can sense if the user has fallen.
- Location Tracking: Adding a GPS module to always know the user's real-time location.
- Local Alerts: A buzzer will provide immediate sound-based warnings to the user.
- Emergency Button: A physical button will be included for the user to manually trigger an alert.
- Web and IoT Integration: We will develop a web application that allows a caregiver to see the user's location and receive status updates, connecting the physical stick to the cloud.

### 3.1.2 Literature Review

Smart assistive technologies have evolved significantly to support the mobility of visually impaired individuals. Among them, smart sticks equipped with ultrasonic sensors, embedded processors, and feedback systems have gained wide attention for their practicality and ease of use.

Recent approaches focus on combining ultrasonic sensing with computer vision to detect obstacles such as stairs, slopes, potholes, and wet floors. Deep learning models like YOLO are used for object recognition, providing real-time audio prompts to guide the user more effectively than traditional cane systems.

To handle uncertain or noisy sensor data, possibilistic modeling has been introduced, allowing reliable stair classification using extracted features from ultrasonic signals. This improves system performance even in real-time embedded setups with limited resources.

In parallel, simpler IoT-based designs emphasize affordability and core functionality. These systems include sensors for obstacle and water detection, GPS for tracking, and GSM modules for emergency alerts, offering essential support in daily navigation without relying on complex algorithms.

Overall, the integration of sensing, processing, and communication technologies is pushing smart stick development toward more intelligent and reliable solutions for visually impaired users.

| Feature                      | Fusion System (2020)    | SVM Ultrasonic (2013)       | Possibilistic Model (2021) | IoT Smart Stick (2021)        |
|------------------------------|-------------------------|-----------------------------|----------------------------|-------------------------------|
| Staircase Detection          | Yes (YOLOv3 + sensors)  | Yes (SVM on ultrasonic FFT) | Yes (Possibility Theory)   | No                            |
| Object Recognition           | Yes (Deep Learning)     | No                          | No                         | No                            |
| Sensors Used                 | Ultrasonic, Camera, Wet | Ultrasonic (1 unit)         | Ultrasonic (1), IMU        | Ultrasonic, IR, Water, SIM808 |
| Feedback                     | Audio                   | -                           | Not specified              | Audio (Buzzer), GSM           |
| ML Method                    | YOLOv3                  | SVM                         | Possibilistic Distribution | Not used                      |
| Accuracy (Stair Recognition) | -                       | ~72.4%                      | Up to 93%                  | Not applicable                |

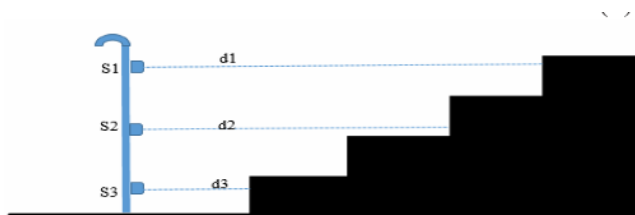


Fig. 4. Stairs obstacle detection

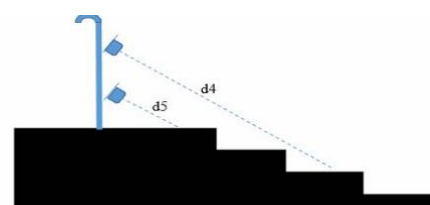


Fig. 7. Downstair obstacle detection

Fig: Previous Works for Stair Case Detection

### 3.1.3 Analysis

Conventional smart sticks for visually impaired individuals primarily use ultrasonic sensors to detect obstacles such as walls, holes, or stairs. However, most existing systems assume that the stick is always held perpendicular or close to the ground, resulting in reduced accuracy when the stick is angled, as is typically the case in real-life usage. In many situations, especially when navigating unfamiliar environments, visually impaired users naturally tilt the stick forward while walking. This leads to incorrect reading or missed detection of critical obstacles like stairs and elevated platforms.

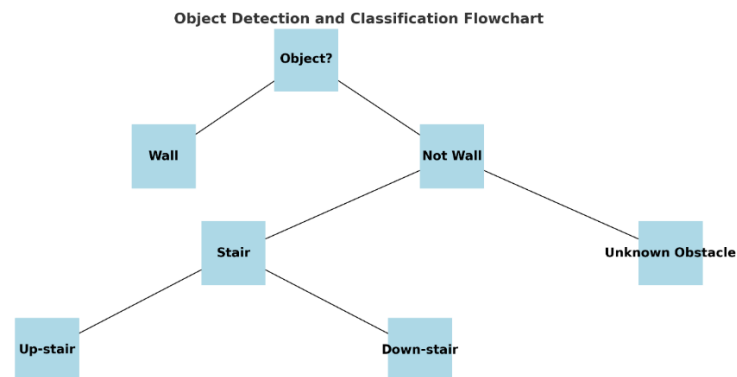
Previous works have proposed systems capable of obstacle detection and even staircase recognition, but their performance is often limited by the placement and angle sensitivity of the sensors. Systems relying heavily on flat-angle sensor orientation struggle to maintain accuracy when the stick is tilted, leading to false positives or unrecognized hazards.

The proposed work addresses this limitation by designing an IoT-based smart stick equipped with ultrasonic sensors strategically positioned to function accurately even when the stick is tilted. This improves the system's ability to differentiate between objects such as **walls, stairs (both ascending and descending), and general obstacles** even if the stick is angled. Additionally, a water sensor ensures safety by detecting slippery surfaces. By preserving accurate detection at different angles, the system aligns more closely with the natural behavior of visually impaired users, enhancing real-world usability and safety.

This approach fills a critical gap in current assistive technologies by improving adaptability and robustness, thereby offering a more realistic and effective solution for independent navigation.

| Feature                             | Existing Smart Sticks                             | Proposed Smart Stick (This Work)                          |
|-------------------------------------|---------------------------------------------------|-----------------------------------------------------------|
| Sensor Type                         | Ultrasonic, sometimes camera or IR                | Ultrasonic + Water sensor+ Accelerometer                  |
| IoT Integration                     | Limited (basic GSM/GPS in some cases)             | Integrated GPS/GSM for tracking and alerts                |
| Obstacle Detection                  | Works well on flat surfaces only                  | Works accurately on flat and inclined surfaces            |
| Stair Detection (Up/Down)           | Inaccurate or inconsistent if stick is angled     | Robust detection even when stick is tilted                |
| Wall/Object Differentiation         | Basic detection; no fine classification           | Differentiates walls, stairs, and other objects           |
| Water Detection (Slippery surfaces) | Often not included                                | Included (using water sensor at stick tip)                |
| Stick Angle Sensitivity             | Requires nearly vertical orientation for accuracy | It can Work if the stick is angled into a definite range. |
| Real-world Usability                | Often limited in dynamic environments             | Designed for natural use posture and diverse environments |

## 3.2 Design Method (PO(a))



### 1. Stair (Upstairs) Detection:

- Upstairs produces a gradual slope in distance:  
 $\text{distance1} > \text{distance2} > \text{distance3}$
- This causes  $df1 = |D1 - D2|$ ,  $df2 = |D2 - D3|$  to be similar.
- If  $|df1 - df2|$  is within a threshold (e.g., 0.5–12 cm), it's likely a staircase.

Key Parameters:

$\text{stairDiff} = \text{abs}(df1 - df2)$

Detected when:  $0.5 < \text{stairDiff} < 12$

### 2. Wall Detection:

- A wall is flat and vertical; all sensors will return similar distances.
- This means  $df1 \approx 0$ ,  $df2 \approx 0$ , and  $\text{stairDiff} \approx 0$ .

Key Parameters:

- Detected when  $df1 < 10$  and  $df2 < 10$  but no significant slope.
- All sensors almost measure similar distances.
- No slope across sensor readings → vertical object (wall).

### 3. Unknown Obstacle Detection:

- If none of the patterns for stairs or walls match, but there's some obstruction (i.e., distance values are small), it's an unknown object.

When triggered:

- When  $\text{stairDiff}$  is outside range, but  $df1$  and  $df2$  are both small.

Diagram Notes:

- Could be a chair leg, bag, or uneven block—non-uniform distances.

### 4. Sensor Arrangement (from diagram)

- Sensor 1 (Top)
- Sensor 2 (Middle)
- Sensor 3 (Bottom)

Sensors are positioned to give a vertical profile of the object ahead. This lets the system capture the shape/structure, whether it's a sloping stair, flat wall, or irregular obstacle.

### 5. Water Detection:

We have used Water sensor for water presence Detection.

## 3.3 Circuit Diagram

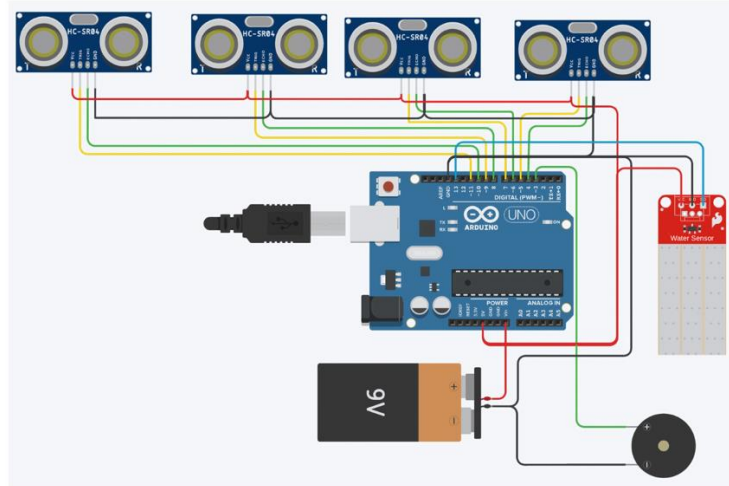


Figure: Circuit Diagram for Object, upstairs, downstairs and water Detection

## 3.4 Hardware Design

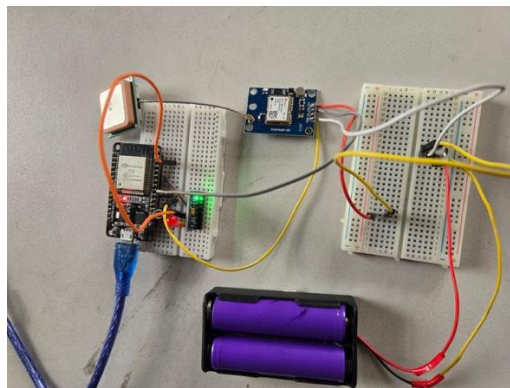
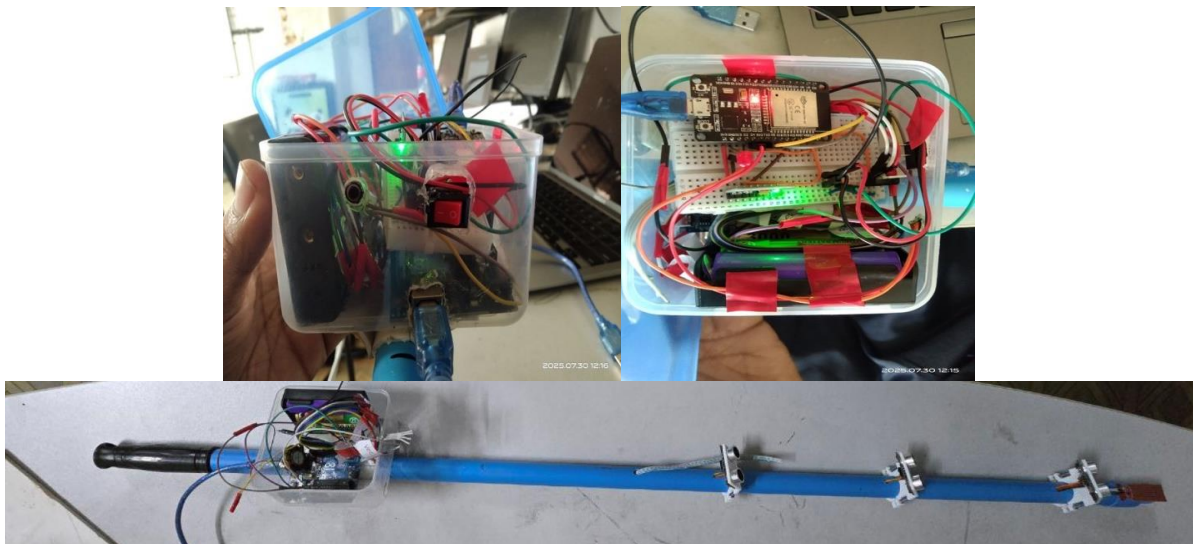


Figure: Circuit for GPS and accelerometer



### 3.5 3D\_ Design



Figure: For Upstair Detection Sonar Holder

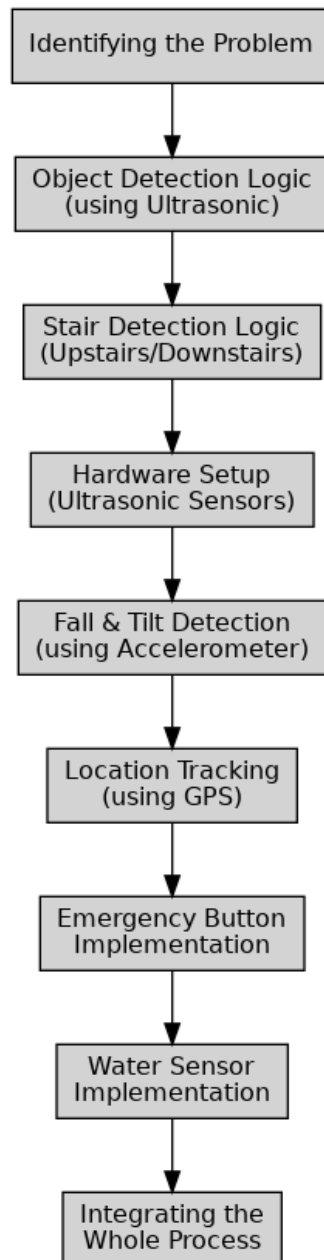


Figure: Down stair Detection Sonar Holder

## 4 Implementation

### 4.1 Description

Final Implementation Overview Flowchart:



The smart blind stick system is designed to enhance mobility and safety for visually impaired individuals. It begins by identifying real-world challenges such as obstacle detection, stair recognition, and fall risk. Ultrasonic sensors are implemented for detecting walls, objects, and stairs—even when the stick is angled. An accelerometer monitors tilt and detects falls. GPS is used for real-time location tracking, while an emergency button allows quick alerts. A water sensor helps identify slippery surfaces. Each module is individually implemented and tested before final integration into a unified system. This comprehensive approach ensures reliable, real-time feedback for safe navigation.

## 4.2 Design Analysis and Evaluation

### 1. Distance Measurement (Ultrasonic Logic):

```
float getDistance(int trigPin, int echoPin) {  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
    long duration = pulseIn(echoPin, HIGH);  
    return duration * 0.0343 / 2.0;  
}
```

- This function sends an ultrasonic pulse and calculates the distance using the time-of-flight of the echo.

Formula:

$$\text{Distance (cm)} = \frac{\text{Duration } (\mu\text{s}) \times 0.0343}{2}$$

Three sensors (d1, d2, d3) measure distance at three different heights.

### 2. Wall Detection:

```
if (df1 < 18 && df2 < 18)
```

- Wall gives **similar** distance readings from all sensors since it's flat and vertical.
- Mathematically:

$$df1 = |d1 - d2|, \quad df2 = |d2 - d3|, \quad \text{and if } |df1 - df2| \approx 0 \Rightarrow \text{Wall}$$

Threshold of 18 cm allows some margin due to angle/tilt of the stick. The wall is detected when the distance readings from all three sensors are nearly equal, indicating a vertical surface. Both  $|d1-d2|$  and  $|d2-d3|$  remain within a close threshold (<18 cm).

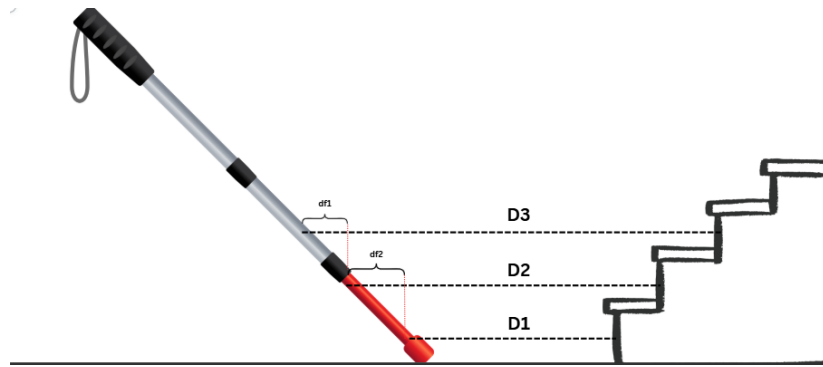
### 3. Staircase Detection(Up-stairs):

```
else if (stairDiff > 0.1 && stairDiff < 12)
```

- A staircase introduces a gradient in readings (like a ramp).
- If the sensors detect gradually decreasing or increasing steps:

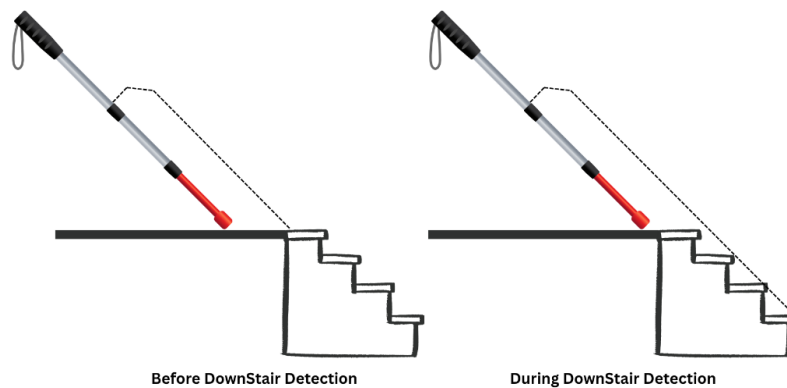
$$df1 = |d1 - d2|, \quad df2 = |d2 - d3|, \quad \text{then } \text{stairDiff} = |df1 - df2|$$

For stairs, stairDiff is small but not zero, typically between 0.1 cm and 12 cm. Staircases are identified by measuring the slope between sensor readings. A consistent gradient across all sensors (non-zero, but small difference between  $|d1-d2|$  and  $|d2-d3|$ ) indicates the presence of stairs.



#### 4. Staircase Detection(down-stairs):

##### DownStair Detection



Before downstairs detection the ultrasonic sensor measures distance of the floor which is fixed. We design the ultrasonic sensor as such that it first detect trade one then trade two.so when there is downstairs the distance increases at fixed distance. Typically the angle of the downward staircase is 30 degree that's why it is very difficult to the trades. We place the ultrasonic sensor at 145 degree with the holder so that the wave from uv sensor detect trades sequentially.

$$d(2) - d(1) = (20-25\text{cm})$$

$$d(3) - d(2) = (20-25\text{cm})$$

here  $d(1)$  is the distance of the floor.  $d(2)$  is the distance of the first trade of downstairs.  $d(3)$  is the distance of the second trade.

4. Manhole Detection: When ultrasonic sensor detects floor it gives a constant distance reading. If the distance increases slowly with constant value (20-25 cm) then it's a down stair. But when the distance increases suddenly it detects a hole in the road.

#### 5. Unknown Obstacle Detection:

```
else {Serial.println(" Type: Unknown Obstacle");
```

##### Explanation:

- If an obstacle is detected (any sensor < threshold), but it's **neither flat** (wall) nor **gradually sloped** (staircase), it's classified as **unknown**.
- Examples: poles, bags, or irregular shapes.

## 6. Fall Detection:

```
float total_acc = sqrt(ax * ax + ay * ay + az * az);

// Free fall detection
if (total_acc < 0.3 && !fallDetected) {
    Serial.println("● FALL DETECTED (FREE FALL)!");
    fallDetected = true;
}
```

- The MPU-6050 measures acceleration along the X, Y, and Z axes.
- The **total acceleration** is calculated using the formula:

$$a_{total} = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

- During **free fall**, the device experiences near-zero net acceleration (i.e.,  $a_{total} < 0.3g$ ), since it's temporarily weightless.
- The system detects this drop and identifies it as a **fall event**.

## 7.Angle Measurement:

```
// Angle with X-axis
float angle_x = acos(ax / total_acc) * 180.0 / PI;

// Angle with Y-axis
float angle_y = acos(ay / total_acc) * 180.0 / PI;
```

- To understand the orientation of the stick, the **angles between the acceleration vector and the X and Y axes** are calculated using:

$$\theta_x = \cos^{-1} \left( \frac{a_x}{a_{total}} \right), \quad \theta_y = \cos^{-1} \left( \frac{a_y}{a_{total}} \right)$$

- These angles indicate the **tilt of the stick relative to the horizontal axes**, useful for further context about its posture before or after a fall.

## WiFi & Web Interface:

- The ESP32 connects to a predefined WiFi network and runs a **local web server** on port 80.
- It hosts a simple **dashboard webpage** that displays:
  - GPS coordinates (Latitude, Longitude)
  - Tilt angles (X and Y) calculated from accelerometer data
- The webpage refreshes every 2 seconds automatically for real-time monitoring.

## GPS Module (TinyGPS++):

- Continuously reads serial data from the GPS module.
- Updates latitude and longitude when a valid location fix is received.

## Web Dashboard UI:

- Designed using minimal **HTML & CSS**.
- Displays values in colored, stylized <div> blocks (.box, .green, .red).
- Helps caregivers access **live location** and **orientation status** remotely.

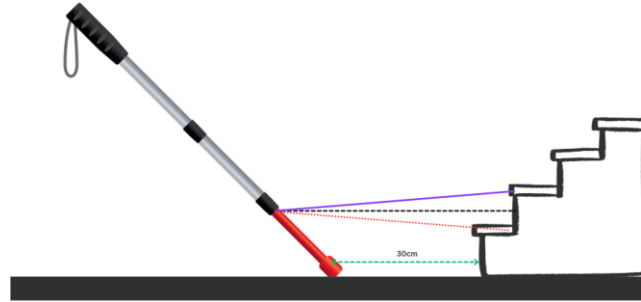
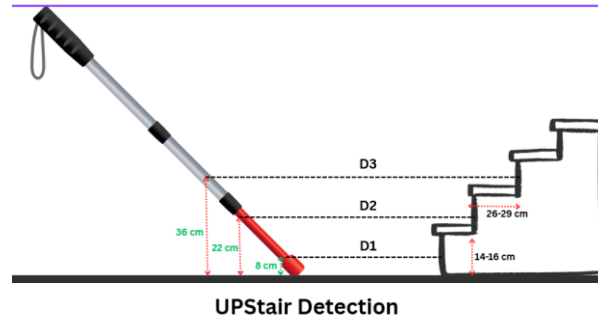
## Emergency Button:

- A **push button** is used to manually trigger an emergency.
- When pressed:

The onboard **LED turns ON**

A message is printed to serial: "🚨 Emergency Button!"

### Mathematical Calculation For Stick Angle:



For  $S_1$ ,

$$\sin \theta = \frac{7}{30} \approx 0.2333 \Rightarrow \theta \approx 13.5^\circ$$

For  $S_2$ ,

$$\sin \theta = \frac{7}{94} \approx 0.0745 \Rightarrow \theta \approx 4.26^\circ$$

For  $S_3$ ,

$$\sin \theta = \frac{7}{58} \approx 0.1207 \Rightarrow \theta \approx 6.93^\circ$$

Note: For more accurate results, we have considered the angle to be  $5^\circ$ .

### 4.3 Novelty:

The key novelty of this project lies in its ability to handle the **inclined orientation** of the stick, which reflects the natural way visually impaired users hold it. Unlike traditional systems that assume vertical sensor alignment, this design uses **trigonometric positioning of ultrasonic sensors** to reliably detect **upstairs and downstairs**, even when the stick is held at an angle. Additionally, the project introduces a **precise and differentiated detection logic** for walls, obstacles, and staircases, enhancing accuracy. A further innovative feature is the integration of a **real-time WiFi dashboard**, allowing caregivers or users to **monitor live data**, including fall events, location, and orientation.

## 4.4 Design Considerations (PO(c))

When we started designing this smart walking stick, we weren't just thinking about circuits and code. We considered the people who would use it and the world they inhabit. Our design choices were guided by three primary principles: prioritizing people's safety, being mindful of our planet's well-being, and respecting the needs of our society.

### 4.4.1 Considerations to public health and safety

This project is all about health and safety. The well-being of the user is our absolute top priority, and every feature is designed to address a real-world risk.

**Preventing Accidents Before They Happen:** The most direct way we protect the user is by preventing falls and collisions. The ultrasonic sensors act as the user's forward-looking eyes, giving them a heads-up about obstacles, walls, or even a sudden downward staircase. This proactive warning system can be the difference between a safe journey and a serious injury.

**Faster Help in an Emergency:** We know that when a fall does happen, every second counts. Our automatic fall detection system is designed to be that crucial first responder.

**Empowering Safe Mobility:** Safety isn't just about preventing physical harm; it's also about mental well-being. The fear of falling or getting lost can cause many elderly and visually impaired individuals to stay home, leading to isolation and a decline in physical health. By giving them a tool that they can trust, we hope to give them the confidence to be more active, mobile.

### 4.4.2 Considerations to environment

**Choosing Rechargeable Power:** Instead of relying on disposable batteries that end up in landfills, we designed the system to use a rechargeable Lithium-ion battery. This makes the device more sustainable for long-term use and significantly reduces waste.

**Energy-Efficient Design:** We specifically chose the ESP32 microcontroller partly for its power-saving features. Our code will be written to make the most of these features, ensuring the stick can run for a long time on a single charge. This not only makes it more convenient for the user but also reduces overall energy consumption.

**Durable and Repairable Components:** We are using standard, durable electronic components. Our goal is to create a device that lasts. By making it from readily available parts, we also open the door for it to be repaired rather than thrown away if something goes wrong, extending the life of the product and reducing e-waste.

### 4.4.3 Considerations to cultural and societal needs

**Promoting Independence and Dignity:** In any culture, personal independence is a source of dignity. Our smart stick is a tool that empowers users to be more self-reliant. It allows them to perform daily activities, like a walk to the local shop or mosque, without needing constant supervision.

**Designing for Inclusivity:** By making it safer for the elderly and visually impaired to navigate public spaces, we are helping to make our society more inclusive. This device encourages participation in community life, reducing the isolation that so many vulnerable

individuals feel. It's a step towards building a community where everyone feels safe and welcome to move around freely.

## 4.5 Investigations (PO(d))

### 4.5.1 Results and Analysis

When water is detected in water sensor & Object type (unknown)

```
D1: 201.00 cm D2: 6.81 cm D3: 39.87
cm
df1 = |D1 - D2|: 195.00 cm df2 = |D2 - D3|: 33.07 cm |df1 - df2|: 161.93
Water Detected!
Obstacle Detected!
Type: Unknown Obstacle
```

No Obstacle only Water Detected:

```
cm
df1 = |D1 - D2|: 590.34 cm df2 = |D2 - D3|: 592.00 cm |df1 - df2|: 1.66
Water Detected!
No Obstacle or Staircase Detected
```

Unknown Obstacle:

```
D1: 199.35 cm D2: 4.20 cm D3: 801.57
cm
df1 = |D1 - D2|: 195.15 cm df2 = |D2 - D3|: 797.37 cm |df1 - df2|: 602.22
Obstacle Detected!
Type: Unknown Obstacle
-----
D1: 199.37 cm D2: 3.65 cm D3: 801.57
cm
df1 = |D1 - D2|: 195.72 cm df2 = |D2 - D3|: 797.92 cm |df1 - df2|: 602.21
Obstacle Detected!
Type: Unknown Obstacle
-----
D1: 199.71 cm D2: 5.27 cm D3: 801.54
cm
df1 = |D1 - D2|: 194.45 cm df2 = |D2 - D3|: 796.27 cm |df1 - df2|: 601.83
Obstacle Detected!
Type: Unknown Obstacle
```

Wall Detection:

```
df1 = |D1 - D2|: 12.62 cm df2 = |D2 - D3|: 16.93 cm |df1 - df2|: 4.30
Obstacle Detected!
Type: Wall
-----
D1: 27.42 cm D2: 39.50 cm D3: 56.15
cm
df1 = |D1 - D2|: 12.07 cm df2 = |D2 - D3|: 16.65 cm |df1 - df2|: 4.58
Obstacle Detected!
Type: Wall
```

Staircase Detection:

```
Obstacle Detected!
Type: Staircase
-----
D1: 8.16 cm D2: 27.53 cm D3: 49.31
cm
df1 = |D1 - D2|: 19.36 cm df2 = |D2 - D3|: 21.78 cm |df1 - df2|: 2.42
Obstacle Detected!
Type: Staircase
```

GPS, Fall Detection and Emergency Push Button:

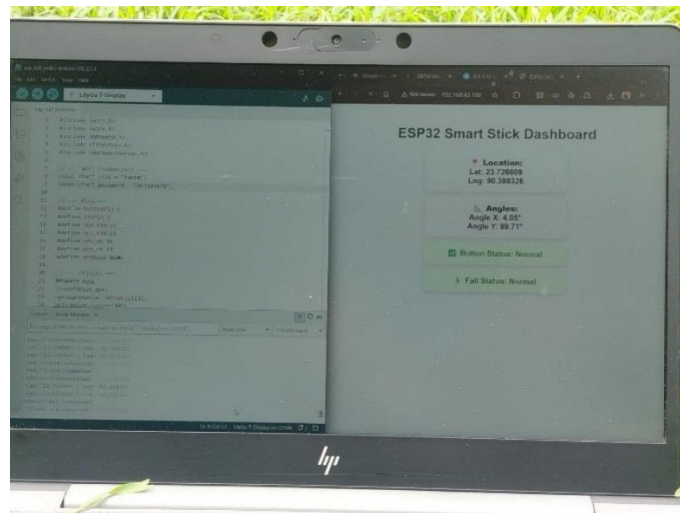


Figure: When No Emergency Button pressed and no fall detected

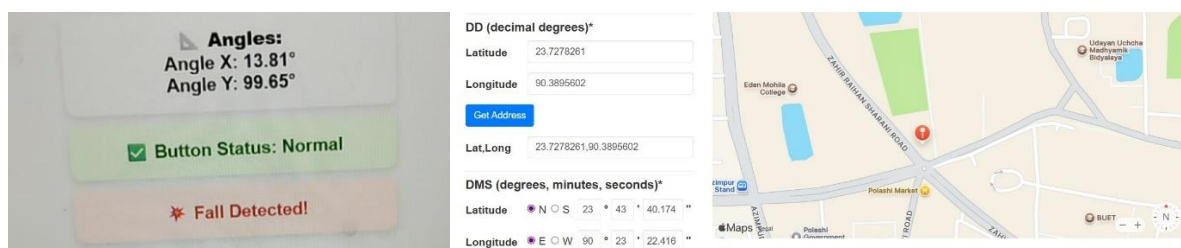


Figure: Fall Detected & Location

## 4.6 Limitations of Tools (PO(e))

Despite its effectiveness, the smart blind stick has several technological limitations. The **ultrasonic sensors** have a limited detection range (typically 2–400 cm) and may produce inaccurate readings on soft, irregular, or angled surfaces, leading to false negatives or unreliable distance data. The detection accuracy is also affected by environmental noise and temperature variations. The **accelerometer** used for fall detection is sensitive to unintentional jerks, which may cause false alarms. **GPS modules** often suffer from poor signal indoors or in densely built areas, limiting real-time tracking accuracy. Additionally, the system requires proper angular alignment; excessive deviation may reduce stair detection performance.

## 4.7 Impact Assessment (PO(f))

As engineers, we have a responsibility to think beyond the technical challenges of a project. We must consider how our work will affect people's lives and the society they live in. This assessment looks at the real-world impact of our smart walking stick, covering the societal, health, safety, and legal dimensions of our design.

### 4.7.1 Assessment of Societal and Cultural Issues

This project has the potential to create a significant positive impact on our society and culture, particularly in how we care for our elders and those with disabilities.

**Promoting Independence and Dignity:** In our culture, we hold our elders in high regard, and their independence is a source of great personal dignity. This smart stick is designed to empower them. By giving users, the confidence to walk to the local market, visit neighbors, or attend prayers at the mosque on their own.

**Strengthening the Family Care Network:** This device is not meant to replace the essential role of family. Instead, it strengthens it. For a son or daughter worried about their elderly parent, the web app provides quiet reassurance. It allows them to check in without being intrusive, respecting the user's independence while providing a vital safety net.

#### 4.7.2 Assessment of Health and Safety Issues

The primary goal of this project is to directly address critical health and safety concerns.

- **A Proactive Approach to Accident Prevention:** The most significant impact is the shift from reactive to proactive safety. A traditional cane only tells you about an obstacle when you hit it. Our smart stick warns you *before* you get there. This advanced warning for walls, stairs, and other hazards is a fundamental improvement that can drastically reduce the number of falls and collisions.
- **Closing the Critical Time Gap in Emergencies:** When a fall occurs, the time it takes to get help can be the difference between a minor incident and a life-altering one. Our system's ability to automatically detect a fall and instantly send an alert with a precise GPS location is arguably its most critical health and safety function.
- **Improving Mental and Physical Well-being:** The constant fear of falling can be a heavy mental burden, leading many to limit their physical activity. By providing a reliable safety tool, we can help reduce this anxiety.

#### 4.7.3 Assessment of Legal Issues

The device will handle sensitive information—the user's real-time location. We must ensure this data is protected. Our system will be designed so that only pre-authorized individuals (e.g., a specific family member) can access the location data. It is crucial to be clear about what this device is and what it is not. It is an *assistive aid*, not a certified medical device. We are not medical professionals, and the device does not guarantee that a user will never fall. This must be communicated clearly to manage expectations and address potential liability concerns. The device is designed to reduce risk, not eliminate it.

Our system is designed to alert caregivers directly via the web app. This is an intentional legal and ethical choice. Triggering a false alarm with official emergency services has serious consequences. By alerting family first, they can assess the situation and make the appropriate call, ensuring that emergency resources are used responsibly.

## 4.8 Sustainability Evaluation (PO(g))

A truly successful engineering solution must be sustainable, meaning it should be beneficial for people and the planet in the long term. We have evaluated our project's sustainability across three key areas.

**Environmental Sustainability:** We believe that technology should not come at the expense of our environment. We are using a rechargeable battery. This single choice prevents hundreds of disposable batteries from ending up in landfills over the device's life. The ESP32 is an energy-efficient microcontroller, and our software will be optimized to use as little power as possible. This means longer battery life and less electricity consumed over time. If a sensor fails, it can be replaced without having to discard the entire unit, which is a key principle in reducing electronic waste

**Social Sustainability:** The project's most significant contribution is to social well-being. The core mission is to provide a safer, more independent life for the elderly and visually impaired. This long-term improvement in their quality of life is the definition of social sustainability.

**Economic Sustainability:** For this device to be truly useful, it must be affordable. We have intentionally selected components that are effective yet inexpensive. As our cost analysis shows, the total cost of the prototype is very reasonable, making it a financially viable solution for many families in Bangladesh and beyond.

## 4.9 Ethical Issues (PO(h))

As the creators of a device designed to help a vulnerable population, we have a profound ethical responsibility to think about the moral implications of our work. Our goal is to do good, but we must also be careful to avoid unintended harm. This means looking beyond the technical challenges to consider the ethical questions our project raises.

At the heart of this project lies a classic ethical dilemma: balancing the need for safety with a person's right to privacy and autonomy. The GPS tracker is a perfect example. While it allows a caregiver to find a loved one in an emergency, it can also feel like surveillance, potentially undermining the user's sense of freedom. Our ethical standpoint is that the user's autonomy is paramount. The device must be a tool for empowerment, not control. If the fall detector doesn't trigger or the GPS gives an incorrect location, the consequences could be severe. This raises a critical ethical question of responsibility. It would be deeply unethical for us to market this stick as a foolproof safety guarantee and have an ethical obligation to consider who will have access to this technology and who will be left out. We intentionally designed the stick to be low-cost by using affordable components. The data our device collects a person's location history is incredibly sensitive. As engineers, we have a moral duty to protect that information. This goes beyond just the legal requirements. It's about earning the trust of the user and their family.

## 5 Reflection on Individual and Team work (PO(i)).

### 5.1 Individual Contribution of Each Member

| Student ID | Contribution                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| 2006127    | Proposal Making, Web development, GPS and Fall detection, Hardware Implementation, Stair Detection, Presentation Making. |
| 2006128    | Stair Detection, GPS and Fall detection, Hardware Implementation, Report writing, Proposal Making                        |
| 2006129    | Proposal Making, Fall detection, Object & Stair detection, Logic Development, Report writing, Presentation Making        |
| 2006130    | Proposal Making, Logistic Supports, Report writing, Presentation Making, Stair Detection                                 |

## 6 Communication to External Stakeholders (PO(j))

### 6.1 Executive Summary

A team of BUET students has developed a smart blind stick to help visually impaired individuals navigate safely and independently. The stick uses ultrasonic sensors to detect obstacles, stairs, and walls, while an accelerometer detects falls. A GPS module tracks location, and an emergency button sends alerts in critical situations. It also includes a water sensor for added safety. Designed to be lightweight and easy to use, the stick provides audio cues for real-time feedback. This innovation combines affordable hardware with intelligent logic, offering an accessible solution that empowers the visually impaired and enhances their mobility and confidence.

### 6.2 User Manual

#### Powering the Stick

- **Turn On:** Press and hold the **power push button**.
- **Turn Off:** Long press the same button again to shut down.
- **Charging:** Plug a standard USB charger into the **charging port**. A small LED indicator shows red while charging and green when fully charged.

#### Sound-Based Alerts

Your stick uses **distinct audio alerts** to inform about the surrounding environment:

| Detection Type         | Sound Indication         |
|------------------------|--------------------------|
| Wall or Obstacle       | Beep-Beep                |
| Upstairs               | Rising Tone Sound        |
| Downstairs             | Falling Tone Sound       |
| Water Detection        | Splash-like Sound        |
| Fall Detected          | Emergency Buzzer Sound   |
| Emergency Button Press | Help Alert in the server |

### Using the Stick:

1. **Hold the stick at a 40° to 50° angle** with the ground for accurate stair detection.
2. As you walk:
  - The stick continuously checks for **walls, stairs, objects, and water** using **ultrasonic sensors**.
  - Alerts are delivered instantly through **audio feedback**.
3. The **accelerometer** tracks the tilt and fall detection. In case of a fall, an emergency alert is triggered.
4. The **GPS module** sends the current location to a pre-set guardian (if connected to a GSM system).
5. **Emergency button** can be pressed in distress to trigger alerts and share location.

## 7 Project Management and Cost Analysis (PO(k))

### 7.1 Bill of Materials for the Prototype

| Component                           | Low Range (BDT) |
|-------------------------------------|-----------------|
| ESP32 Microcontroller               | 450             |
| Accelerometer (MPU6050)             | 150             |
| Ultrasonic Sensors (HC-SR04) x4     | 200             |
| Li-ion or Li-polymer Battery (3.7V) | 80              |
| Water Sensor                        | 60              |
| Battery Management Circuit (TP4056) | 50              |
| Buzzer                              | 20              |
| Wires & Connectors                  | 150             |
| Prototyping Board                   | 100             |
| Resistors, Capacitors, Transistors  | 50              |
| Switches/Buttons                    | 10              |
| Arduino UNO                         | 400             |
| <b>Total</b>                        | <b>1890</b>     |

## 7.2 Bulk Production Cost Estimation

| Component                           | Prototype Cost (BDT) | Estimated Bulk Cost (BDT) | Notes                                                                 |
|-------------------------------------|----------------------|---------------------------|-----------------------------------------------------------------------|
| ESP32 Microcontroller               | 450                  | 200                       | Significant price reduction through bulk purchasing from suppliers.   |
| Accelerometer (MPU6050)             | 150                  | 50                        | Price can be reduced substantially.                                   |
| Ultrasonic Sensors (HC-SR04) x4     | 200                  | 100                       | Sensors are inexpensive in bulk.                                      |
| Li-ion or Li-polymer Battery        | 80                   | 60                        | Battery costs can be slightly reduced.                                |
| Water Sensor                        | 60                   | 30                        | Small, low-cost sensor with good bulk pricing.                        |
| Battery Management Circuit (TP4056) | 50                   | 30                        | Simple circuit, good price reduction potential.                       |
| Buzzer                              | 20                   | 10                        | Very inexpensive in bulk.                                             |
| Wires & Connectors                  | 150                  | 50                        | Bulk wire and connector purchases are significantly cheaper.          |
| Prototyping Board                   | 100                  | 0                         | Replaced by a custom-designed PCB in bulk production.                 |
| Resistors, Capacitors, Transistors  | 50                   | 20                        | Discrete components are extremely cheap in bulk.                      |
| Switches/Buttons                    | 10                   | 5                         | Minor cost component, but still reducible.                            |
| Arduino UNO                         | 400                  | 0                         | Replaced by the more cost-effective ESP32 used for the final product. |
| <b>Subtotal</b>                     | <b>1820</b>          | <b>555</b>                |                                                                       |
| <b>Additional Costs</b>             |                      |                           |                                                                       |
| Custom PCB                          | 0                    | 50                        | Cost of designing and manufacturing a custom printed circuit board.   |
| Enclosure and Housing               | 0                    | 200                       | Cost of the physical casing for the device.                           |
| Assembly and Labor                  | 0                    | 100                       | Labor costs for assembling each unit.                                 |
| <b>Total Bulk Production Cost</b>   | <b>1820</b>          | <b>905</b>                |                                                                       |

Based on these estimates, the projected bulk production cost is approximately **BDT 905** per unit, a significant reduction from the prototype cost of BDT 1890.

### Cost Reduction Opportunities

The following strategies can be implemented to further reduce the bulk production cost:

- Custom PCB Integration:** The current prototype uses an Arduino UNO and an ESP32. In bulk production, these would be replaced by a single, custom-designed Printed Circuit Board (PCB) using the ESP32 as the main microcontroller. This eliminates the cost of the Arduino UNO and the prototyping board, while also reducing the number of wires and connectors required. This is a major area of savings.

- **Component Sourcing:** Sourcing components directly from manufacturers or large-scale distributors (e.g., AliExpress, Alibaba for small-scale, or local/international wholesalers for larger scale) can yield much lower prices than retail. Negotiating bulk contracts will be key.
- **Simplifying the Design:** The BOM lists both an ESP32 and an Arduino UNO. It's likely the Arduino UNO was used for initial prototyping. The final product should use only the more powerful and cost-effective ESP32, which will also handle all the necessary sensor inputs.
- **Alternative Sensors:** While HC-SR04 sensors are cheap, for a final product, more compact and reliable ultrasonic sensors could be considered. However, given their low cost, it's unlikely to provide significant savings unless a single, higher-quality sensor could replace multiple ones.
- **Enclosure Design:** Optimizing the enclosure design for injection molding or other high-volume manufacturing techniques can significantly lower the cost per unit compared to 3D printing or other prototyping methods. Using standard, off-the-shelf enclosures could also reduce costs.
- **Integrated Power Management:** Instead of a separate Battery Management Circuit (e.g., TP4056 module), the power management functionality can be directly integrated into the custom PCB design, potentially reducing component count and assembly time.

### 7.3 Timeline of Project Implementation

| Phase                                     | Start Week | End Week |
|-------------------------------------------|------------|----------|
| Phase 1: Research & Requirement Analysis  | 5          | 6        |
| Phase 2: Logic Design & Trial             | 6          | 8        |
| Phase 3: Object Detection                 | 8          | 9        |
| Phase 4: Stair Detection                  | 9          | 10       |
| Phase 5: Web App Development              | 10         | 10       |
| Phase 6: GPS, Fall Detection              | 10         | 11       |
| Phase 6: Hardware Implementation          | 11         | 13       |
| Phase 6: Final Deployment & Documentation | 13         | 14       |

## 8 Future Work (PO(I))

- **Adaptive Angle Detection**  
Integrate an angle correction algorithm using real-time IMU (gyroscope + accelerometer) data so the system automatically adjusts to changes in stick orientation, enhancing accuracy in stair detection.
- **Weight Optimization**  
Use compact PCBs, lighter materials (e.g., carbon fiber or lightweight polymers), and integrated modules to make the stick more comfortable for long-term use.
- **Voice Assistance with AI**

Incorporate offline voice feedback using pre-trained models to alert users in their native language about the detected object type (e.g., “Wall ahead,” “Go upstairs”).

➤ **Obstacle Memory Mapping**

Use GPS and timestamp data to mark frequently encountered obstacles and stairs on a basic map, allowing route customization for the user.

➤ **Object Classification with ML**

Upgrade to a camera or LIDAR-based solution and apply basic machine learning to identify objects beyond just stairs or walls (e.g., furniture, doors, people).

➤ **Waterproofing & Weather Resistance**

Improve durability by sealing electronics with waterproof enclosures and adding protection against heat, dust, and rain.

## 9 References

• **Fusion of Object Recognition and Obstacle Detection Approach for Assisting Visually Challenged Person**

*IEEE*, 2020

This paper presents a smart stick with ultrasonic sensors and object recognition techniques to help visually impaired individuals detect various types of obstacles.

<https://ieeexplore.ieee.org/document/9260062>

• **An Embedded System for Staircase Detection to Assist Visually Impaired People**

*IEEE*, 2016

Discusses the use of embedded sensors for detecting staircases (upstairs/downstairs) and provides effective algorithms for real-time recognition.

<https://ieeexplore.ieee.org/document/7480160>

• **Smart Stick for Blind People Using Ultrasonic Sensors and GPS Module**

*International Journal of Engineering Research & Technology (IJERT)*, 2017

A basic implementation of object detection, navigation, and GPS integration for outdoor mobility.

<https://www.ijert.org/research/smart-stick-for-blind-people-using-ultrasonic-sensors-and-gps-module-IJERTV6IS050409.pdf>

• **Design and Implementation of a Smart Cane for Visually Impaired People**

*International Journal of Scientific & Engineering Research*, 2019

Includes water sensing, ultrasonic obstacle detection, and buzzer feedback — closely aligned with your stick’s functionalities.

<https://www.ijser.org/researchpaper/Design-and-Implementation-of-a-Smart-Cane-for-Visually-Impaired-People.pdf>

• **IoT Based Fall Detection and Location Tracking System for Elderly**

*IEEE*, 2021

Although elderly-focused, this paper gives useful insight into accelerometer-based fall detection integrated with GPS and emergency alert systems.

<https://ieeexplore.ieee.org/document/9440601>

• **Design of an Assistive Device for Visually Impaired People Using Raspberry Pi and Ultrasonic Sensor**

*International Journal of Advanced Research in Computer Engineering & Technology*

(IJARCET), 2018

A Raspberry Pi-based implementation that performs wall and pit detection using multiple ultrasonic sensors.

[https://scholar.google.com/scholar\\_lookup?title=Design%20of%20an%20Assistive%20Dev  
ice%20for%20Visually%20Impaired...](https://scholar.google.com/scholar_lookup?title=Design%20of%20an%20Assistive%20Device%20for%20Visually%20Impaired...)

## 10 Full Code:

```
//Code For Obstacle, Wall, Staircase & Water Detection System

// === Pin Definitions ===
const int trigPin1 = 4;
const int echoPin1 = 5;
const int trigPin2 = 6;
const int echoPin2 = 7;
const int trigPin3 = 8;
const int echoPin3 = 9;

const int ledPin = 13;
const int speakerPin = 10; // Speaker connected to pin 10
const int waterPin = 11; // Water sensor input

// Threshold
const float obstacleThreshold = 30.0;

// Detection history buffer (0 = none, 1 = wall, 2 = staircase, 3 = unknown, 4 = water)
int detectionHistory[3] = {0,0, 0};

void setup() {
  Serial.begin(9600);

  pinMode(trigPin1, OUTPUT); pinMode(echoPin1, INPUT);
  pinMode(trigPin2, OUTPUT); pinMode(echoPin2, INPUT);
  pinMode(trigPin3, OUTPUT); pinMode(echoPin3, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(speakerPin, OUTPUT);
  pinMode(waterPin, INPUT);

  Serial.println("Ⓜ Obstacle, Wall, Staircase & Water Detection System Ready!");
  Serial.println("-----");
}

float getDistance(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH);
  return duration * 0.0343 / 2.0;
}
```

```

}

void playSound(int type) {
  switch (type) {
    case 1: tone(speakerPin, 1000, 500); break; // Wall
    case 2: tone(speakerPin, 1500, 500); break; // Staircase
    case 3: tone(speakerPin, 2000, 500); break; // Unknown
    case 4: tone(speakerPin, 2500, 500); break; // Water Detected
    default: noTone(speakerPin); break;
  }
  delay(600);
}

void loop() {
  float d1 = getDistance(trigPin1, echoPin1); delay(50);
  float d2 = getDistance(trigPin2, echoPin2); delay(50);
  float d3 = getDistance(trigPin3, echoPin3);

  float df1 = abs(d1 - d2);
  float df2 = abs(d2 - d3);
  float stairDiff = abs(df1 - df2);

  Serial.print("D1: "); Serial.print(d1); Serial.print(" cm\t");
  Serial.print("D2: "); Serial.print(d2); Serial.print(" cm\t");
  Serial.print("D3: "); Serial.println(d3); Serial.println(" cm");

  Serial.print("df1 = |D1 - D2|: "); Serial.print(df1); Serial.print(" cm\t");
  Serial.print("df2 = |D2 - D3|: "); Serial.print(df2); Serial.print(" cm\t");
  Serial.print("|df1 - df2|: "); Serial.println(stairDiff);

  // === Water Sensor Detection ===
  if (digitalRead(waterPin) == HIGH) {
    Serial.println("💧 Water Detected!");
    digitalWrite(ledPin, HIGH);
    playSound(4);
    delay(1000);
    digitalWrite(ledPin, LOW);
  }

  bool isObstacle = (d1 <= obstacleThreshold || d2 <= obstacleThreshold || d3 <= obstacleThreshold);
  int detectionType = 0;

  if (isObstacle) {
    Serial.println("🟡 Obstacle Detected!");

    if (df1 < 18 && df2 < 18) {
      Serial.println("📦 Type: Wall");
      digitalWrite(ledPin, HIGH);
      detectionType = 1;
    }
    else if (stairDiff > 0.1 && stairDiff < 12) {
      Serial.println("🪜 Type: Staircase");
      digitalWrite(ledPin, HIGH);
    }
  }
}

```

```

        detectionType = 2;
    }
    else {
        Serial.println("△ Type: Unknown Obstacle");
        digitalWrite(ledPin, HIGH);
        detectionType = 3;
    }
} else {
    Serial.println("□ No Obstacle or Staircase Detected");
    digitalWrite(ledPin, LOW);
    noTone(speakerPin);
}

// === Update detection history ===
detectionHistory[0] = detectionHistory[1];
detectionHistory[1] = detectionHistory[2];
detectionHistory[2] = detectionType;

if (detectionHistory[0] == detectionType &&
    detectionHistory[1] == detectionType &&
    detectionHistory[2] == detectionType &&
    detectionType != 0) {
    playSound(detectionType);
    detectionHistory[0] = detectionHistory[1] = detectionHistory[2] = 0;
}

Serial.println("-----");
delay(500);
}

```

#### Code for Location Tracking, Angle Detection, and Web-output:

```

#include <WiFi.h>
#include <Wire.h>
#include <MPU6050.h>
#include <TinyGPS++.h>
#include <HardwareSerial.h>

// === WiFi Credentials ===
const char* ssid = "Redmi Note 13";
const char* password = "12345678";

// === Pins ===
#define buttonPin 4
#define ledPin 2
#define SDA_PIN 21
#define SCL_PIN 22
#define GPS_RX 16
#define GPS_TX 17
#define GPSPBaud 9600

MPU6050 mpu;
TinyGPSPlus gps;
HardwareSerial GPSSerial(2);

```

```

WiFiServer server(80);

bool emergencyTriggered = false;
unsigned long emergencyStartTime = 0;
bool fallDetected = false;
unsigned long fallDetectedTime = 0;
int buttonState = HIGH;
String latitude = "Waiting...";
String longitude = "Waiting...";

// === FUNCTION: Ensure WiFi is connected ===
void ensureWiFiConnected() {
  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("WiFi Disconnected. Reconnecting...");
    WiFi.begin(ssid, password);
    int retries = 0;
    while (WiFi.status() != WL_CONNECTED && retries < 20) {
      delay(500);
      Serial.print(".");
      retries++;
    }
    if (WiFi.status() == WL_CONNECTED) {
      Serial.println("\nReconnected! IP: " + WiFi.localIP().toString());
    } else {
      Serial.println("\nFailed to reconnect.");
    }
  }
}

void setup() {
  Serial.begin(115200);
  GPSSerial.begin(GPSBaud, SERIAL_8N1, GPS_RX, GPS_TX);
  Wire.begin(SDA_PIN, SCL_PIN);

  pinMode(buttonPin, INPUT_PULLUP);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);

  Serial.print("Connecting to WiFi...");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000); Serial.print(".");
  }
  Serial.println("\nWiFi connected!");
  Serial.print("IP Address: "); Serial.println(WiFi.localIP());

  server.begin();

  Serial.println("Initializing MPU6050...");
  mpu.initialize();
  if (!mpu.testConnection()) {
    Serial.println("MPU6050 connection failed!");
    while (1);
  }
}

```

```

}
Serial.println("✅ MPU6050 Ready");
}

void loop() {
  ensureWiFiConnected(); // <<<<<< CHECK FOR WIFI EVERY LOOP

  while (GPSserial.available()) {
    gps.encode(GPSserial.read());
    if (gps.location.isUpdated()) {
      latitude = String(gps.location.lat(), 6);
      longitude = String(gps.location.lng(), 6);
    }
  }

  buttonState = digitalRead(buttonPin);
  if (buttonState == LOW && !emergencyTriggered) {
    emergencyTriggered = true;
    emergencyStartTime = millis();
    digitalWrite(ledPin, HIGH);
  }
  if (emergencyTriggered && millis() - emergencyStartTime >= 5000) {
    emergencyTriggered = false;
    digitalWrite(ledPin, LOW);
  }

  int16_t ax_raw, ay_raw, az_raw;
  mpu.getAcceleration(&ax_raw, &ay_raw, &az_raw);
  float ax = ax_raw / 16384.0;
  float ay = ay_raw / 16384.0;
  float az = az_raw / 16384.0;
  float total_acc = sqrt(ax * ax + ay * ay + az * az);

  if (total_acc < 0.8 && !fallDetected) {
    fallDetected = true;
    fallDetectedTime = millis();
  }
  if (fallDetected && millis() - fallDetectedTime >= 5000) {
    fallDetected = false;
  }

  float angle_x = 0, angle_y = 0;
  if (total_acc != 0) {
    angle_x = acos(ax / total_acc) * 180.0 / PI;
    angle_y = acos(ay / total_acc) * 180.0 / PI;
  }

  WiFiClient client = server.available();
  if (client) {
    String request = client.readStringUntil('\r');
    client.flush();

    String html = "<!DOCTYPE html><html><head><meta charset='UTF-8'>";

```

```

html += "<meta http-equiv='refresh' content='2'/>";
html += "<title>IoT-Based Smart Walking Stick</title>"; // Changed title
html += "<style>";
html += "body{font-family:sans-serif;text-align:center;background:#f0f2f5;padding:20px;}";
html += "h1, h2, h3, h4{color:#2c3e50; margin-bottom: 5px;}"; // Added styles for H2, H3, H4
html += ".box{border-radius:10px;padding:15px;margin:10px auto;width:300px;}";
html += "box-shadow:0 4px 8px rgba(0,0,0,0.1);font-size:18px;font-weight:bold;}";
html += ".green{background:#e6ffe6;color:#28a745;} .red{background:#ffe6e6;color:#dc3545;}";
html += "</style></head><body>";

// New header content
html += "<h2>EEE 416 (January 2025)</h2>";
html += "<h3>Microprocessor and Embedded Systems Laboratory</h3>";
html += "<h4>Final Project Report</h4>";
html += "<h4>Section: B2 Group: 08</h4>";
html += "<h1>IoT-Based Smart Walking Stick for Enhanced<br>Mobility of the Elderly and Visually Impaired  
People.</h1>"; // Main title

html += "<div class='box'><b>📍 Location:</b><br>Lat: " + latitude + "<br>Lng: " + longitude + "</div>";
html += "<div class='box'><b>📐 Angles:</b><br>Angle X: " + String(angle_x, 2) + "°<br>Angle Y: " +  
String(angle_y, 2) + "°</div>";

html += emergencyTriggered
    ? "<div class='box red'>🚨 Emergency Help Needed!</div>"
    : "<div class='box green'>✅ Emergency Help Status: Normal</div>";

html += fallDetected
    ? "<div class='box red'>🚨 Fall Detected! PLease Help!</div>"
    : "<div class='box green'>🧎 Fall Status: Normal</div>";

html += "</body></html>";

client.println("HTTP/1.1 200 OK");
client.println("Content-type:text/html");
client.println();
client.println(html);
client.stop();
}

delay(50);
}

```